

Lecture 6 – Solar Thermal Energy

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ECE180J



Outline

- Heat Transfer Mechanisms
- Passive Solar Heating
- Active Solar Heating

- B3 Chap 3; B1 Chap 4, 8.1-8.2
- Abdul Hai Alami etc, "Concentrating solar power (CSP) technologies: Status and analysis," International Journal of Thermofluids, Volume 18, 2023.

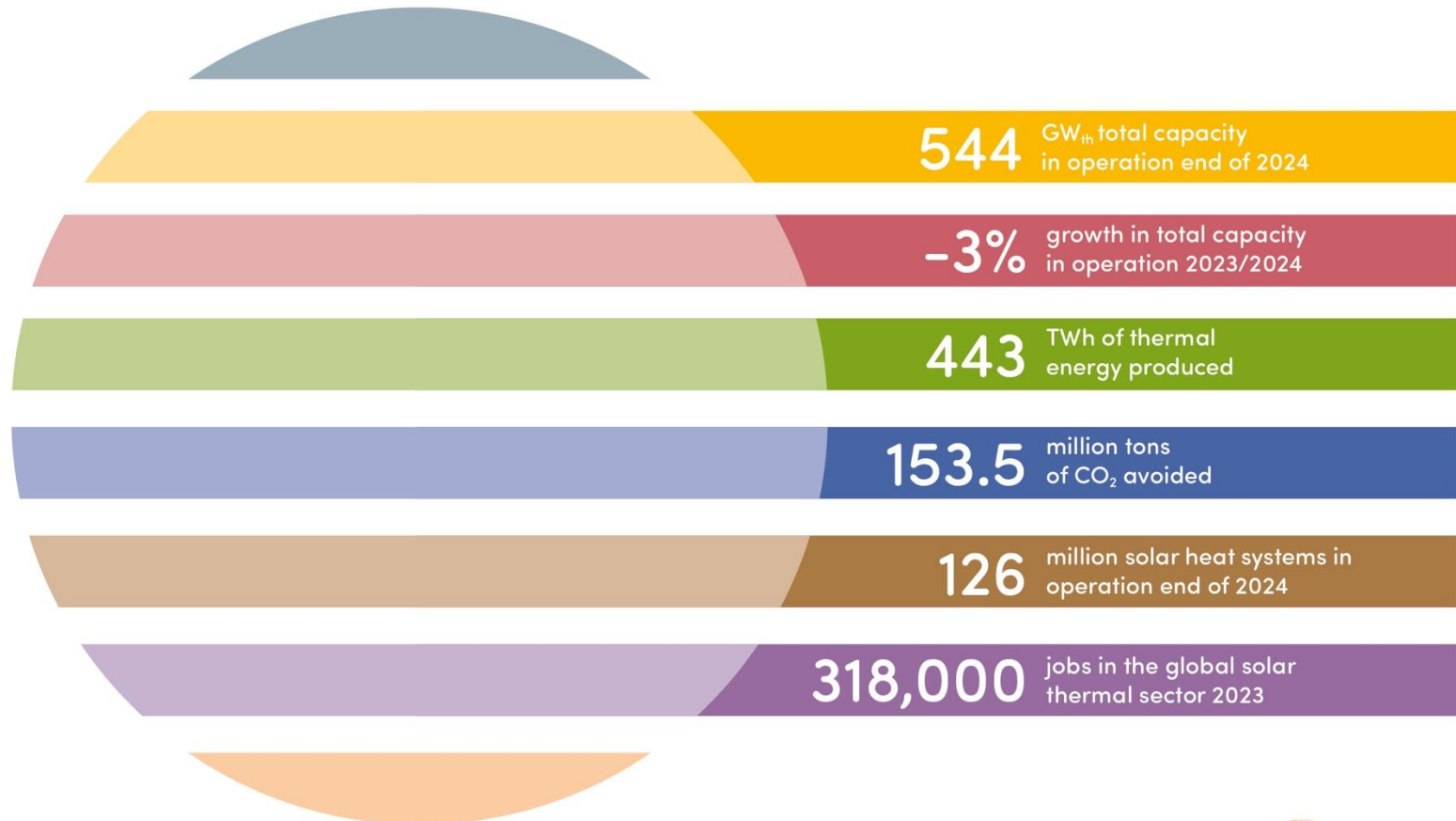
Solar Thermal Energy

- [Solar thermal energy](#) is a form of energy and a technology for harnessing solar energy to generate thermal/electrical energy.
- Solar thermal technologies can produce heat for **hot water, space heating and industrial processes**, with systems ranging from small residential scale to very large community and industrial scale.
- The required temperature to meet the heat demand determines the **collector type and design**.



Solar Heat Worldwide 2025

Solar heat on the global scale

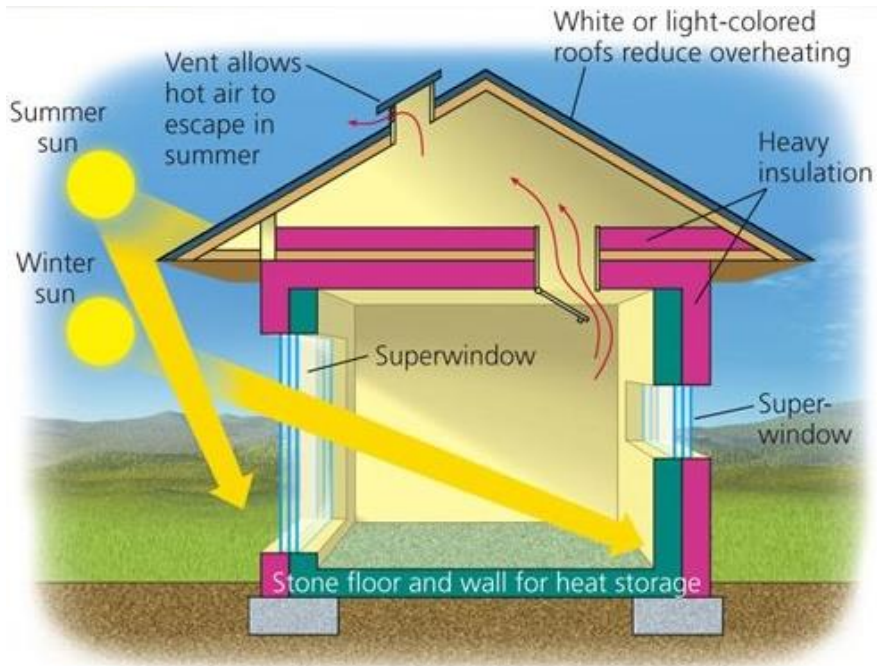


IEA Solar Heating & Cooling Programme | www.iea-shc.org/solar-heat-worldwide



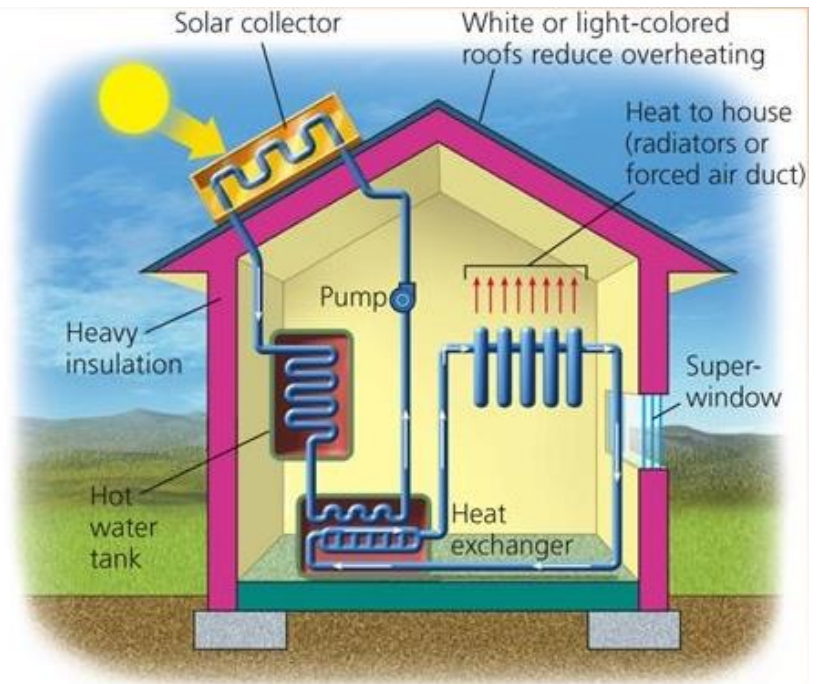
Active Heating vs Passive Heating

Passive solar heating:
absorb solar energy
directly into a building



PASSIVE

Active solar heating:
involves a discrete solar collector
to gather solar radiation

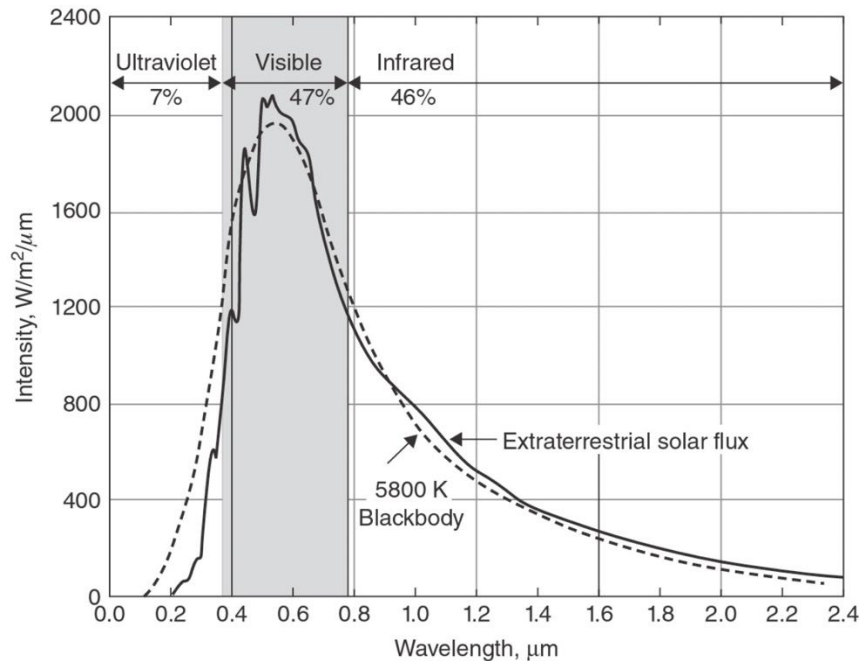


ACTIVE

- Cold air is denser than warm air (molecules closer together)
- More water vapor → lower air density

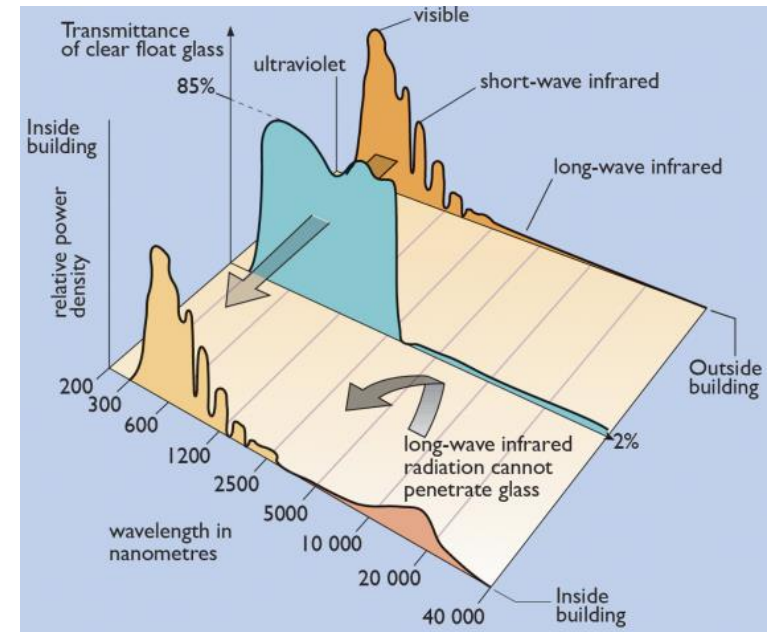
Passive Solar Heating

Glass as Filter



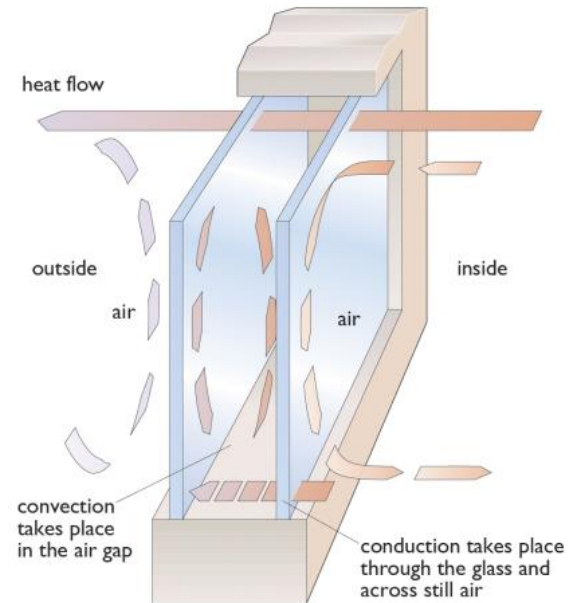
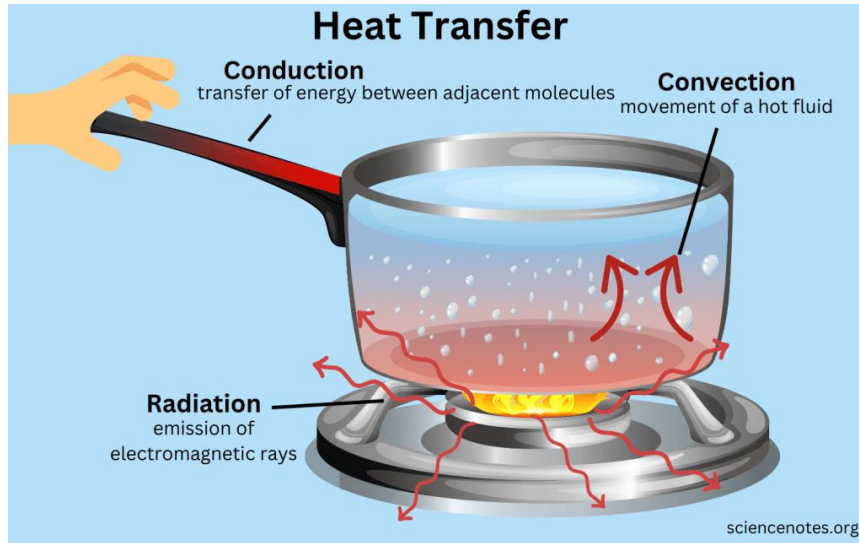
Most low-temperature solar collection is dependent on the properties of **glass**:

- Transparent to visible light & short-wave infrared radiation
- Opaque to long-wave infrared



Make glass transparent → maximize its **transmittance** (*fraction of incident light passing through it*) → minimize the iron content of the glass

Heat Loss Mechanisms

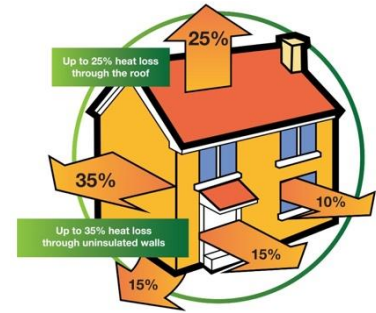


- **Conduction:** transfer of heat through direct molecular interaction and vibration, **without any bulk motion** of the substances involved.
- **Convection:** transfer of **heat by the actual bulk movement and circulation of a substance (liquid or gas)** due to density differences caused by temperature variations.
- **Radiation:** transfer of heat in the form of electromagnetic waves, which can travel through a vacuum or transparent medium without relying on any intervening substance.

Heat Flow Rate Equation

We're interested in regulating the rate at which the thermal energy is transferred. For example:

- keep an object at a temperature different from that of its surrounding for a long time by slowing down heat flow.
- want an object to cool down rapidly by increasing the rate at which thermal energy is transferred.

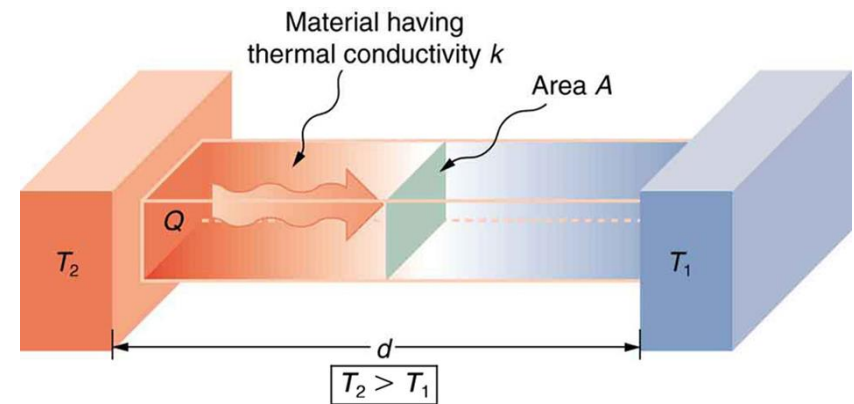


Law of heat conduction:

$$\frac{\Delta Q}{\Delta t} = -\lambda A \frac{\Delta T}{d}$$

Where:

- $\frac{\Delta Q}{\Delta t}$: heat transfer rate
- k : thermal conductivity
- A : cross-sectional area
- ΔT : temperature difference
- d : material thickness



The minus sign indicates that heat flows opposite to the temperature gradient ($\frac{\Delta T}{d}$), i.e., from high to low temperature.

Thermal Conductivity

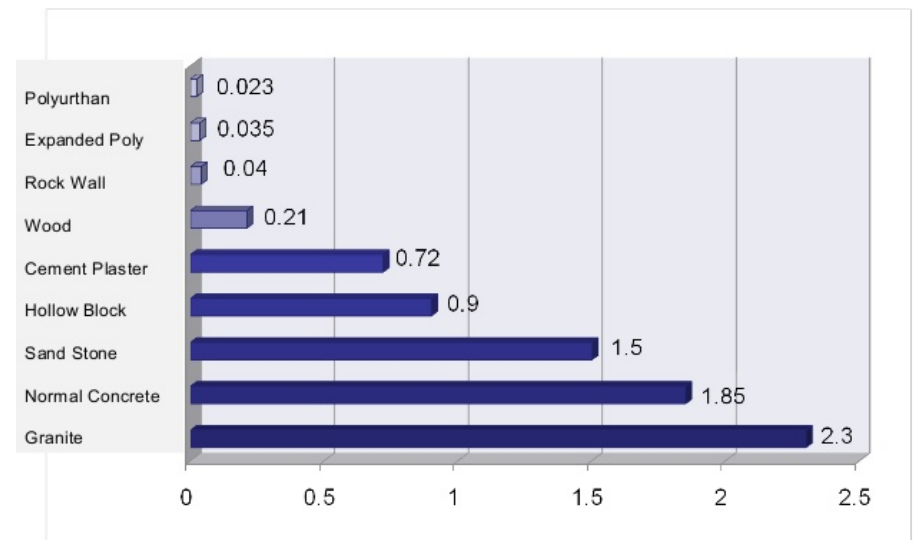
The unit of thermal conductivity factor is: Watt/(meter × Kelvin)

Temperature conversions

	from Celsius	to Celsius
Fahrenheit	$[^{\circ}\text{F}] = [^{\circ}\text{C}] \times \frac{9}{5} + 32$	$[^{\circ}\text{C}] = ([^{\circ}\text{F}] - 32) \times \frac{5}{9}$
Kelvin	$[\text{K}] = [^{\circ}\text{C}] + 273.15$	$[^{\circ}\text{C}] = [\text{K}] - 273.15$

Material	Thermal conductivity (W/m K)
Copper (pure)	399
Gold (pure)	317
Aluminum (pure)	237
Iron (pure)	80.2
Carbon steel (1 %)	43
Stainless Steel (18/8)	15.1
Glass	0.81
Plastics	0.2 – 0.3
Wood (shredded/cemented)	0.087
Cork	0.039
Water (liquid)	0.6
Ethylene glycol (liquid)	0.26
Hydrogen (gas)	0.18
Benzene (liquid)	0.159
Air	0.026

Thermal Conductivity Comparison $W/(m.K)$



Example of Heat Flow Equation

Heat flow equation:

$$\frac{\Delta Q}{\Delta t} = -\lambda A \frac{\Delta T}{d}$$

Q: Find the rate of heat flow transferred through a 40 cm long aluminum bar with a cross section area of 16 cm². The thermal conductivity of the bar is $\lambda = 205 \text{ W}/(\text{m} \cdot \text{K})$. The temperature at the hot end is 120°F while at the cold end is 20°F.

A: The key is to correctly convert some units.

$$120^\circ\text{F} = 322.039^\circ\text{K}, 20^\circ\text{F} = 266.483^\circ\text{K} \implies \Delta T = 55.56^\circ\text{K}$$

$$\frac{\Delta Q}{\Delta t} = \lambda A \frac{\Delta T}{d} = 205 \times 16 \times 10^{-4} \times \frac{55.56}{0.4} = 45.56 \text{ W}$$

Window U-value and Heat Loss

Heat flow rate per square meter = U-value $\times$ temperature difference

Table 3.3 Indicative U-values for windows with wood or PVC-U frames

Glazing type	W m ⁻² K ⁻¹
Single glazing	4.8
Double glazing (normal glass, air filled)	2.7
Double glazing (hard coat low-e, emissivity = 0.15, air filled)	2.0
Double glazing (hard coat low-e, emissivity = 0.2, argon filled)	2.0
Double glazing (soft coat low-e, emissivity = 0.05, argon filled)	1.7
Triple glazing (soft coat low-e, emissivity = 0.05, argon filled)	1.3

Source: BRE, 2013

The lower the U-value, the better the insulation performance!

Quiz

Q1: What is the rate of heat loss through a large single-glazed window with an area of 2m^2 , on a day when indoor-outdoor temperature difference is 15°C ?
What is the total heat loss on that day?

$$\begin{aligned} \text{A: Heat loss rate} &= 2 \times 4.8 \times 15 = 144 \text{ W} \Rightarrow \\ \text{Total heat loss} &= 144 \times 24 = 3.456 \text{ kWh} \end{aligned}$$

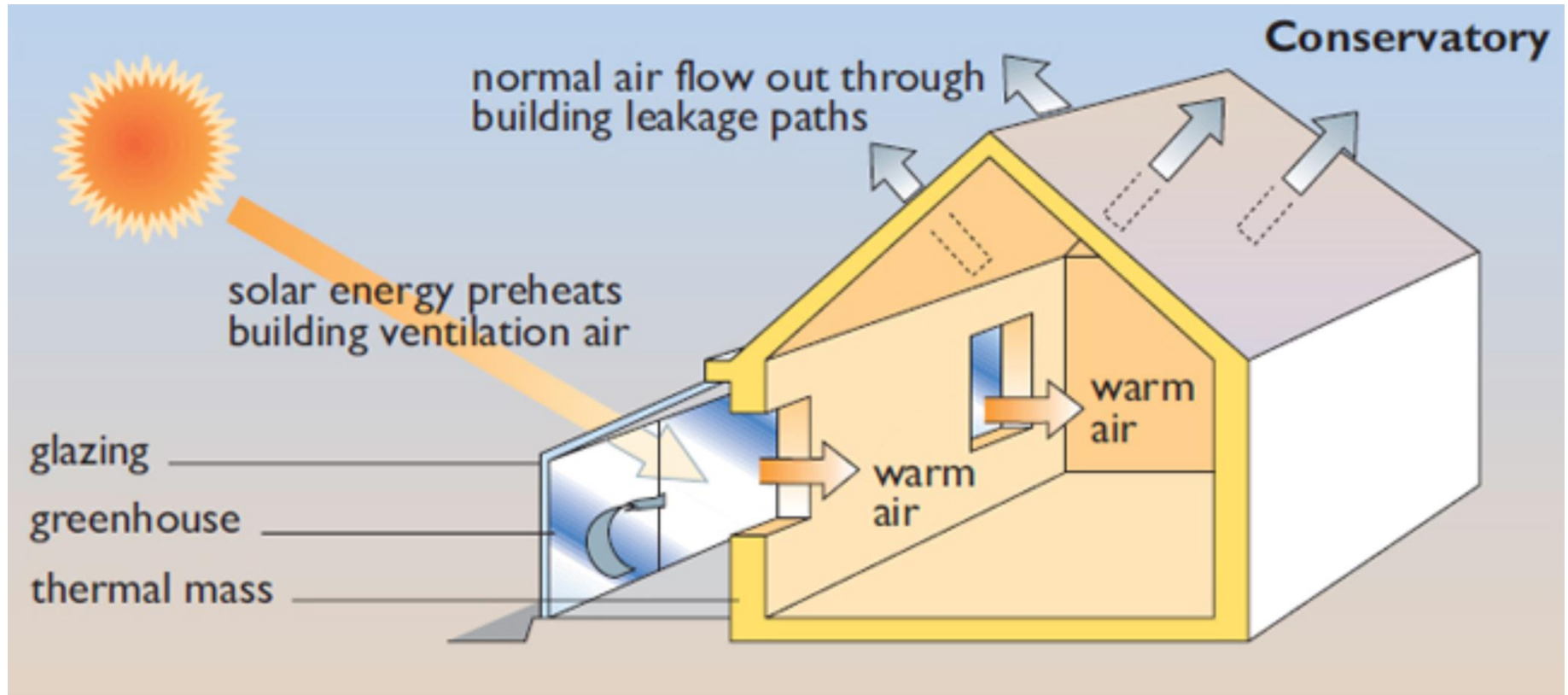
Q2: What is the rate of heat loss through a large single-glazed window with an area of 2m^2 , on a day when indoor-outdoor temperature difference is 15°F ?
What is the total heat loss on that day?

$$\begin{aligned} \text{A : } [^\circ\text{F}] &= [^\circ\text{C}] \times \frac{9}{5} + 32 \Rightarrow \Delta T_{[^\circ\text{F}]} = \Delta T_{[^\circ\text{C}]} \times \frac{9}{5} = \Delta T_{[K]} \times \frac{9}{5} \\ &\Rightarrow \Delta T_{[K]} = \Delta T_{[^\circ\text{F}]} \times \frac{5}{9} \end{aligned}$$

$$\text{Heat loss rate} = 2 \times 4.8 \times 15 \times \frac{5}{9} = 80\text{W} \Rightarrow$$

$$\text{Total heat loss} = \text{Heat loss rate} \times \text{time duration} = 80 \times 24 = 1.92\text{kWh}$$

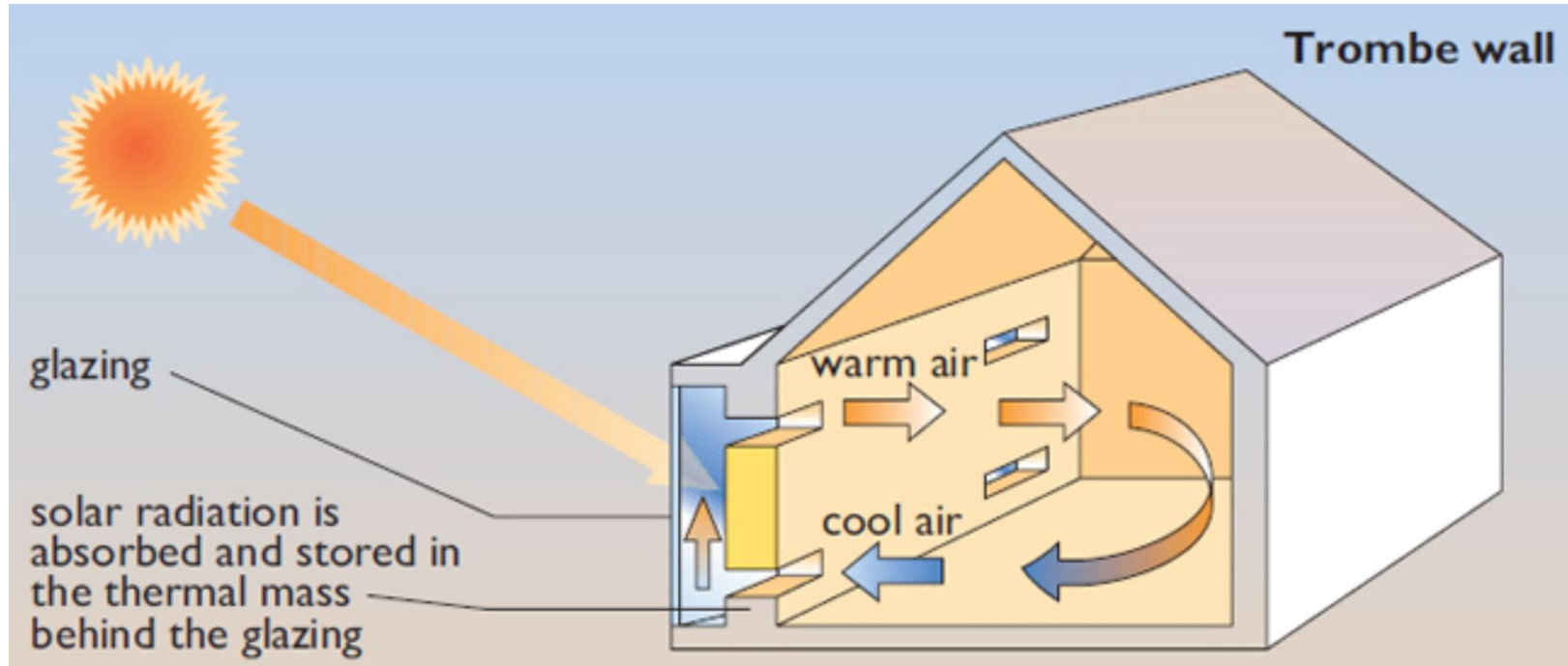
Passive Solar Heating – Conservatory



A conservatory/greenhouse:

The south side of a building can be considered as a kind of habitable solar collector where air is the heat transfer fluid, carrying energy into the building behind, and the energy store is the building itself, especially the wall at the back.

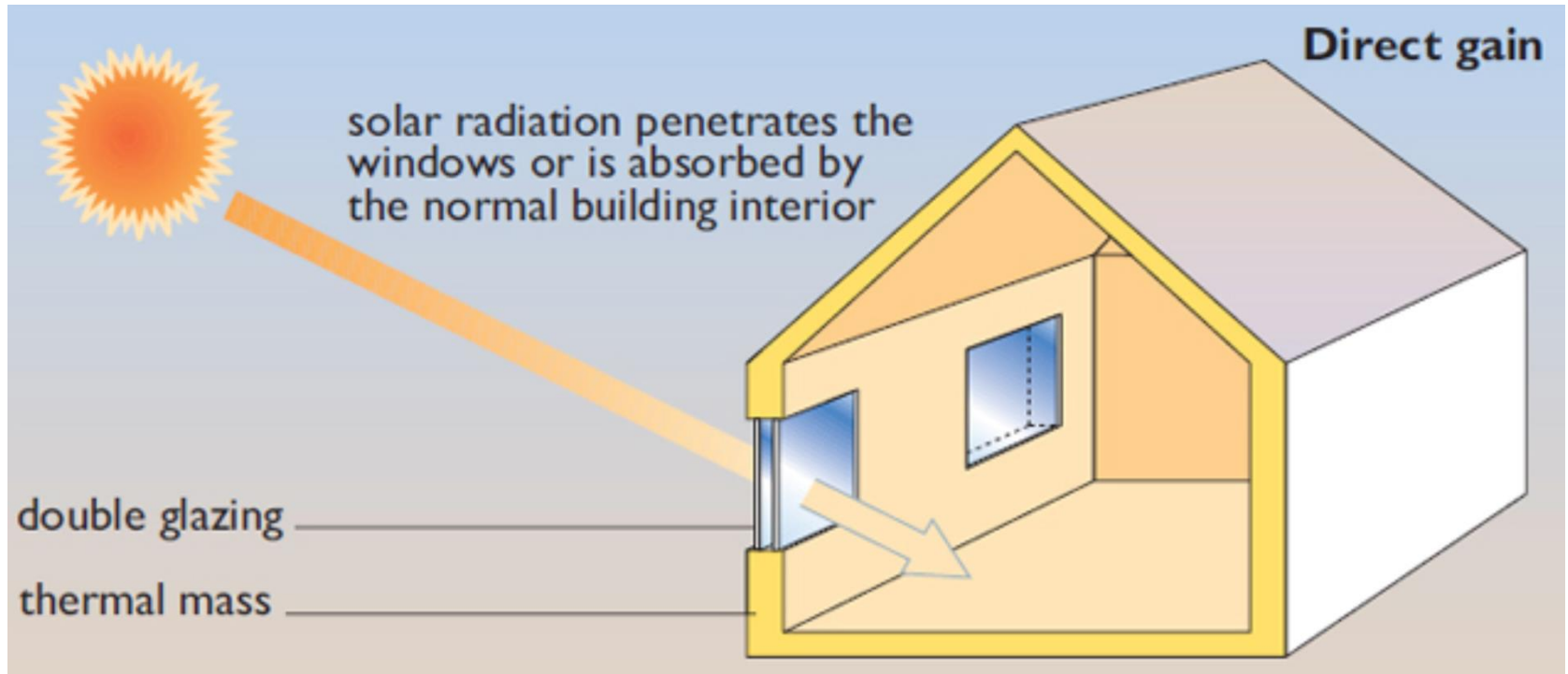
Passive Solar Heating – Trombe Wall



A Trombe wall (named after its French inventor, Félix Trombe):

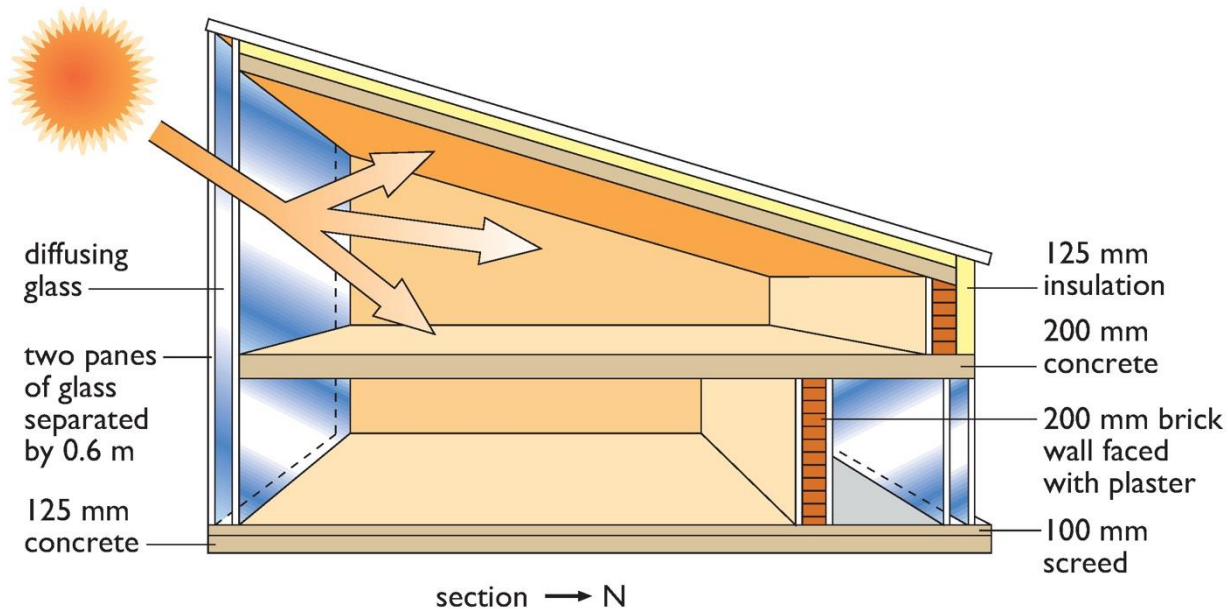
- Conservatory is replaced by a thin glazed air space in front of a storage wall.
- Solar radiation warms the storage wall and is radiated into the house from its inner side.
- On Sunny days air is circulated through vents in the air space into the house behind. At night and on cold days, this air flow is cut off.

Passive Solar Heating – Direct Gain



- All glazed buildings make use of 'direct gains' of energy from the Sun to some degree.
- If the building is 'thermally massive' enough, i.e. built of heavy materials such as concrete, and the heating system is responsive, the solar heat gains are useful.
- If the building is too 'thermally lightweight', such as one of timber frame construction, it may overheat on Sunny days.

Wallasey Solar Campus

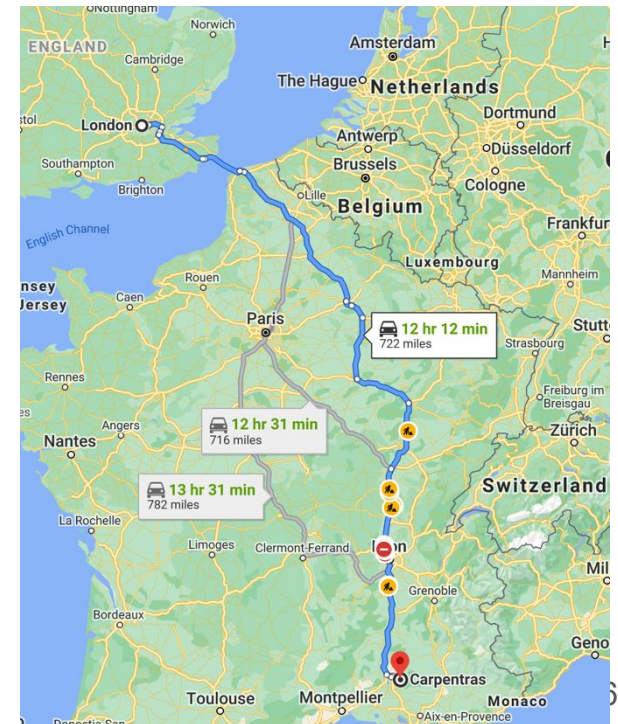
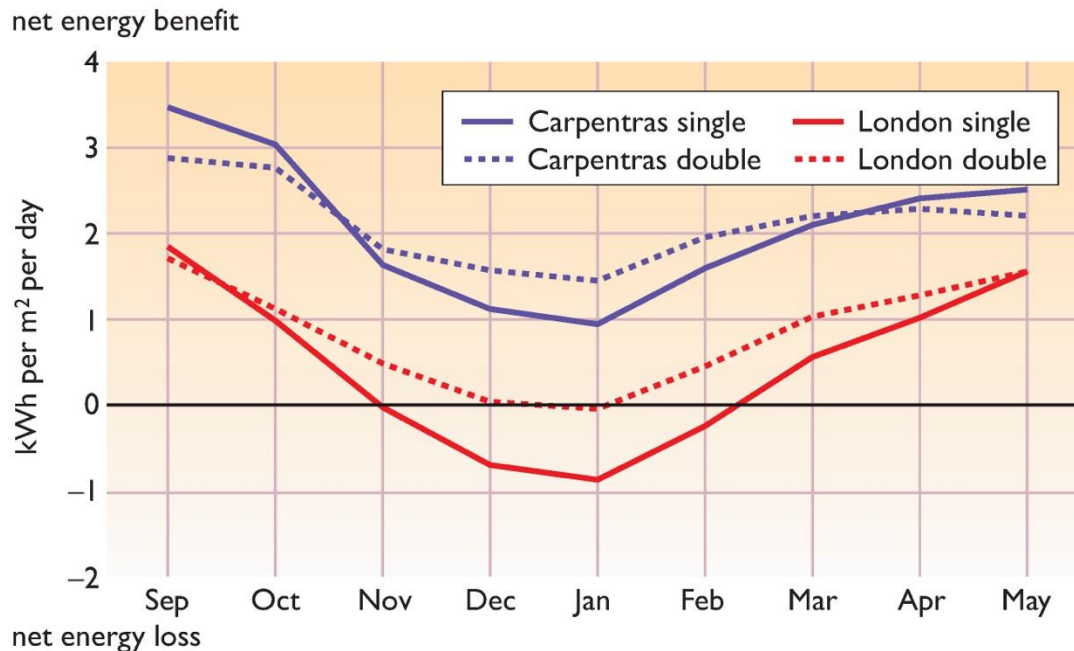


Window Energy Balance

Q: Can a window gets more heat flows in than out so that it provides net energy benefits? What factors shall we consider?

A: Depends on the following factors:

- 1) average internal and external temperature
- 2) available solar radiation
- 3) transmittance, U-value, orientation, and shading of the window



Quiz

Q: What general guidelines can you think of for optimizing the use of passive solar heating in buildings?

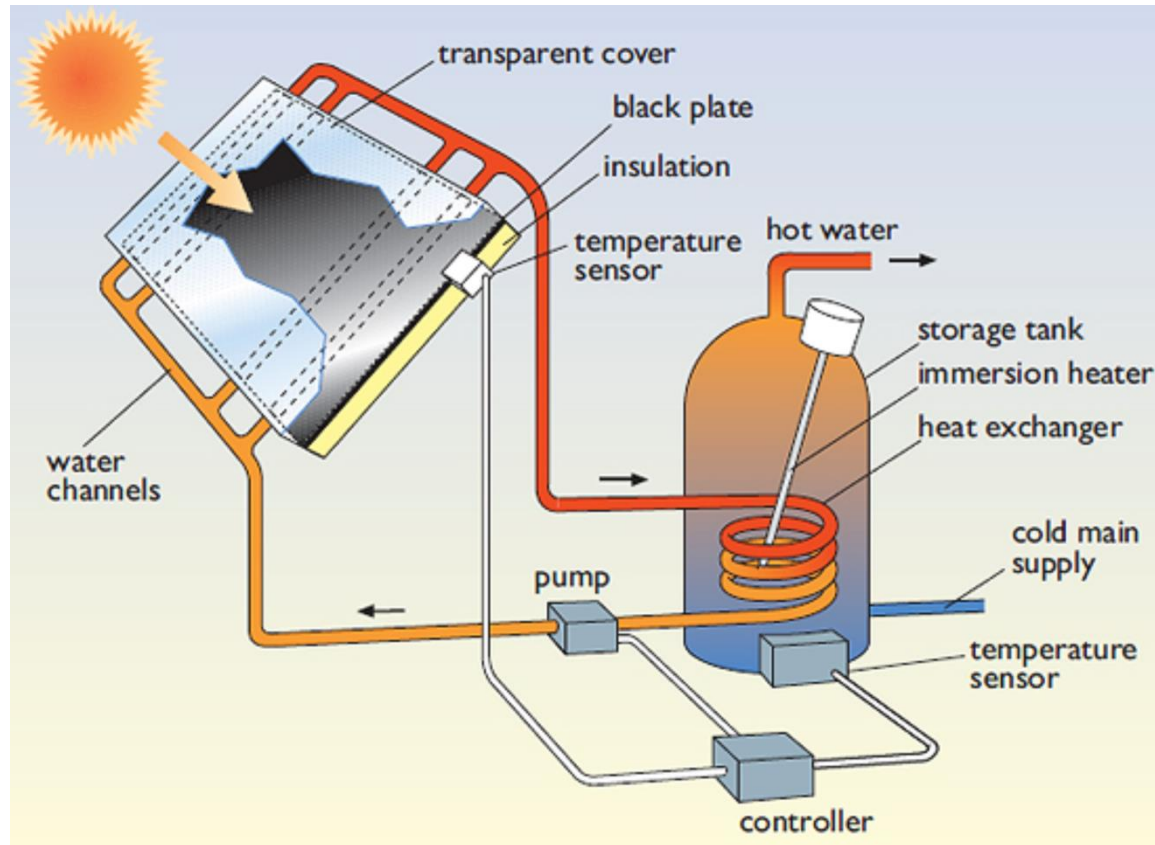
Answer:

- High insulation to minimize heat loss
- Efficient backup heating system
- South-facing design (SE–SW); concentrate glazing and living spaces on south side
- Avoid winter shading from nearby buildings/trees
- Include thermal mass to store heat and reduce summer overheating

Active Solar Heating

Pumped Solar Water Heater

1. A collector panel, consisting of glazing, an absorber plate, and insulation
2. An insulated hot water storage tank, with hot water from the panel circulating through a heat exchanger situated at the bottom
3. A pumped circulation system containing an anti-freeze transferring the heat from the panel to the store



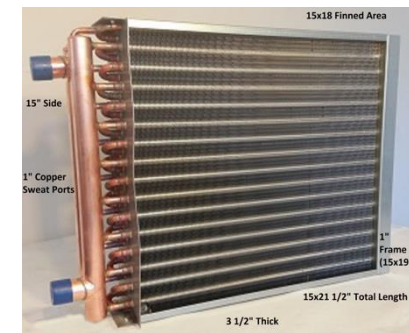
Heater Exchanger

Heater exchanger: A system used to transfer heat between two or more fluids.

- The fluids may be separated by a solid wall to prevent mixing or they may be in direct contact.
- Heat exchangers are widely used in both cooling and heating processes: space heating, refrigeration, air conditioning, power stations, chemical plants, petrochemical plants, petroleum refineries, natural-gas processing, and sewage treatment.



Shell and Tube Heat Exchanger



Water to Air Heat Exchanger

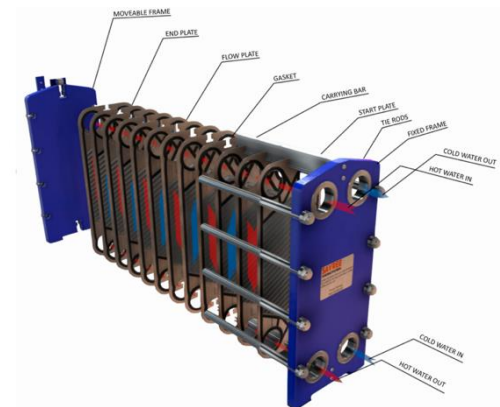
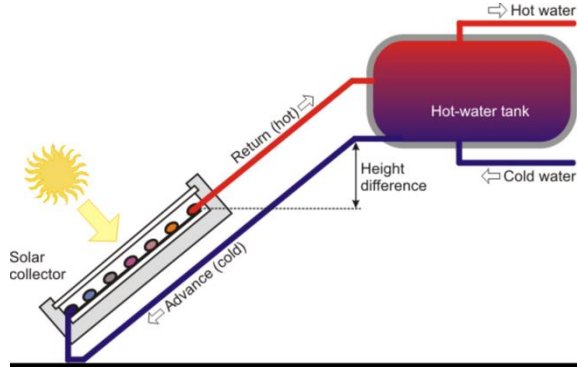


Plate Heat Exchanger

Thermosyphon Solar Water Heater



Consists of a solar collector and an elevated storage tank:

- Sunlight heats the water in the collector; as it warms, it becomes less dense and naturally rises into the tank.
- Cooler water from the tank flows down to the collector, creating a continuous circulation loop.

- No pump is required (driven purely by **natural convection**).
- Typically includes an electric backup heater for cloudy periods. In many designs, domestic water flows directly through the collector (no heat exchanger).

Key Difference from Regular (Active) Solar Water Heaters

- *Thermosyphon*: passive circulation (no pump/controller/electricity needed), simpler, lower cost, but requires tank placement above collector.
- *Regular/Active systems*: use pumps and controllers, allow flexible installation, often include heat exchangers for indirect heating.

History of Thermosyphon Water Heater

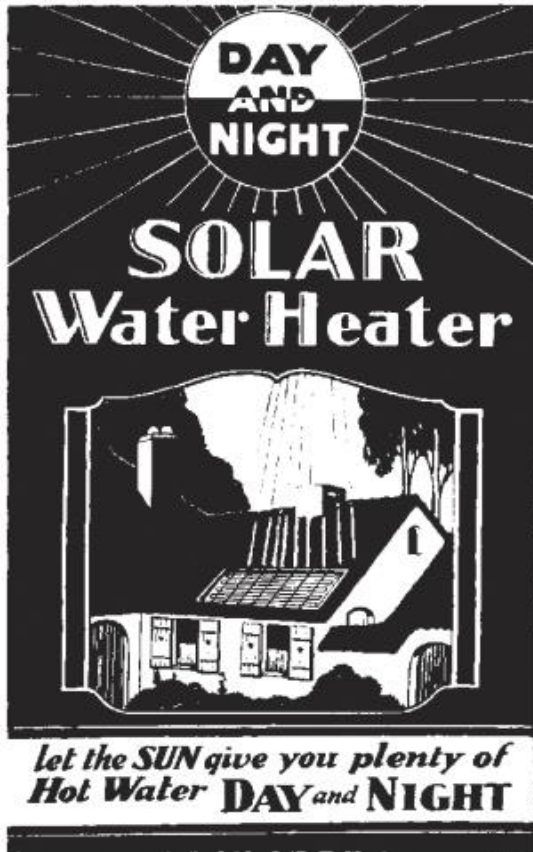


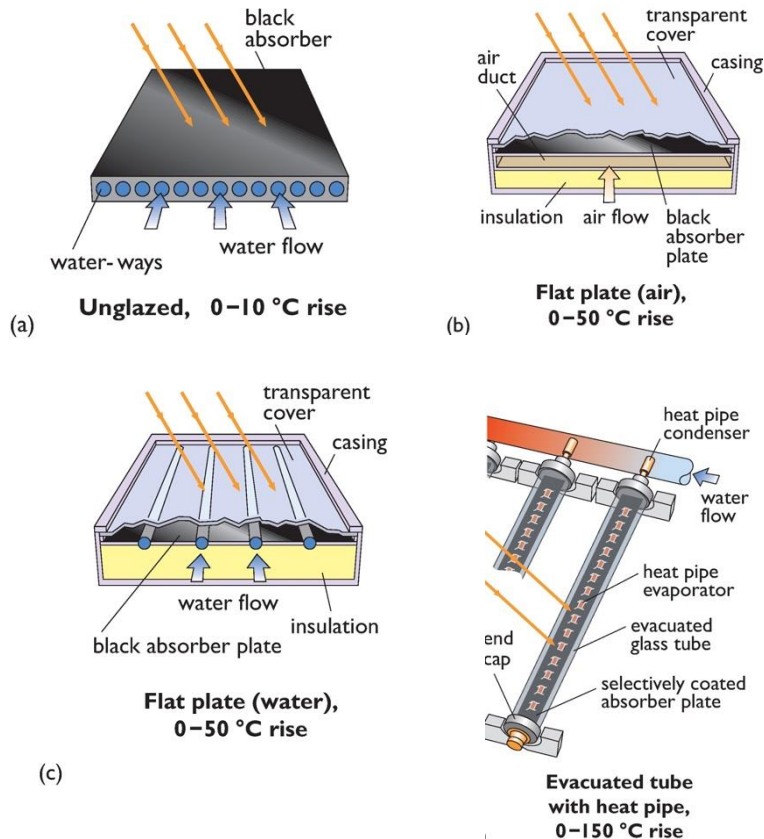
Figure: An advertisement for Bailey's solar water heaters in 1915.

- Thermosyphon water heater was patented in 1909 by William J. Bailey
- The system had an insulated tank, which could keep water hot overnight. Bailey called his business the 'Day and Night' Solar Water Heater Company
- Sold about 4000 systems before the local discovery of cheap natural gas in the 1920s that virtually closed his business



William Bailey

Varieties of Solar Collectors



Water is a better heat transfer fluid than air because it can absorb and transfer more heat per unit volume and per unit time than air, due to higher specific heat capacity and thermal conductivity

(a) suitable for swimming pool heating, heat losses are relatively unimportant.

(b) not so common, mainly used for applications such as crop drying.

(c) mainstay of domestic solar water heating usually single glazed. With a more elaborate glazing system a higher temperature difference can be sustained.

(d) the absorber plate is a metal strip running down the center of each tube:

- heat losses due to convection are suppressed by a vacuum in the tube
- lower heat losses allow a better performance in winter, but more expensive

Solar District Heating



Sunmark A/S, Denmark

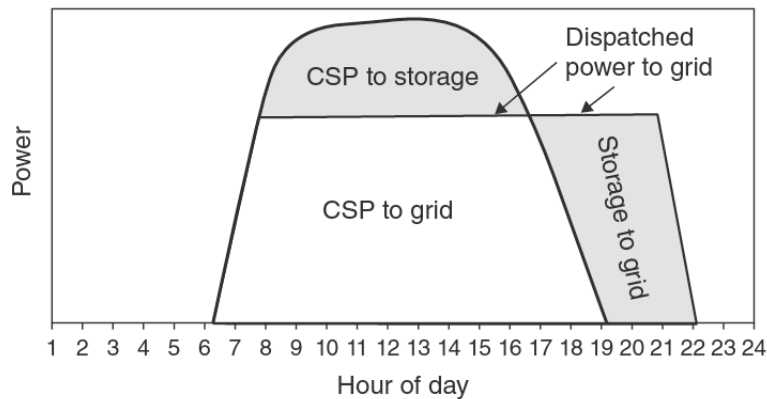
18,000 m² array of collectors in Marstal, Denmark.
A further 15,000 m² array and a 75,000 m³ pit heat store
have been stored to the left of this picture.

This supplies 50% of heating for 1650 households.

Concentrating Solar Power (CSP)

CSP technologies convert sunlight into thermal energy to run a heat engine to power a generator.

CSP can be used either as a *flexible provider of electricity* or *fossil-fuel backup or hybridization* that allows existing fossil fuel projects to run cleaner.

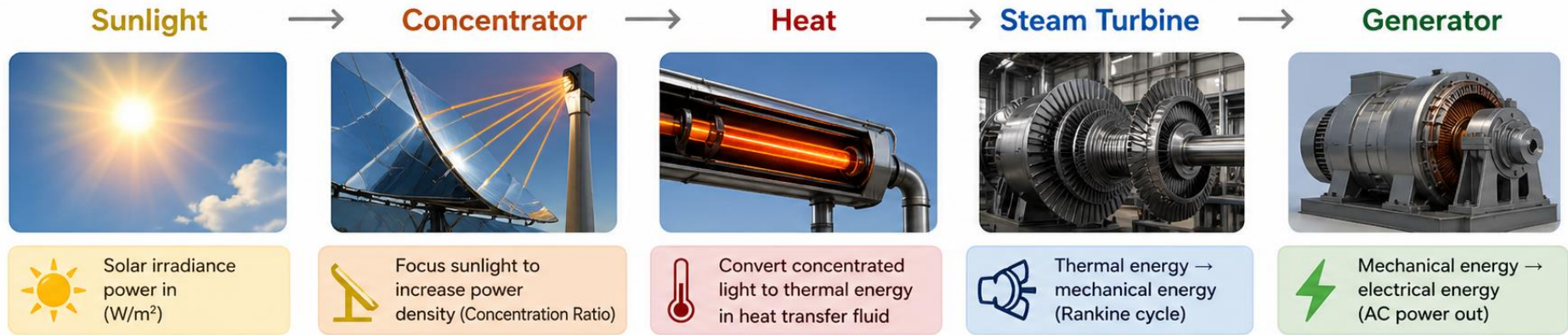


(a) Ivanpah Solar Power Facility (CA)



(b) SEGS (CA)

CSP Energy Conversion Chain



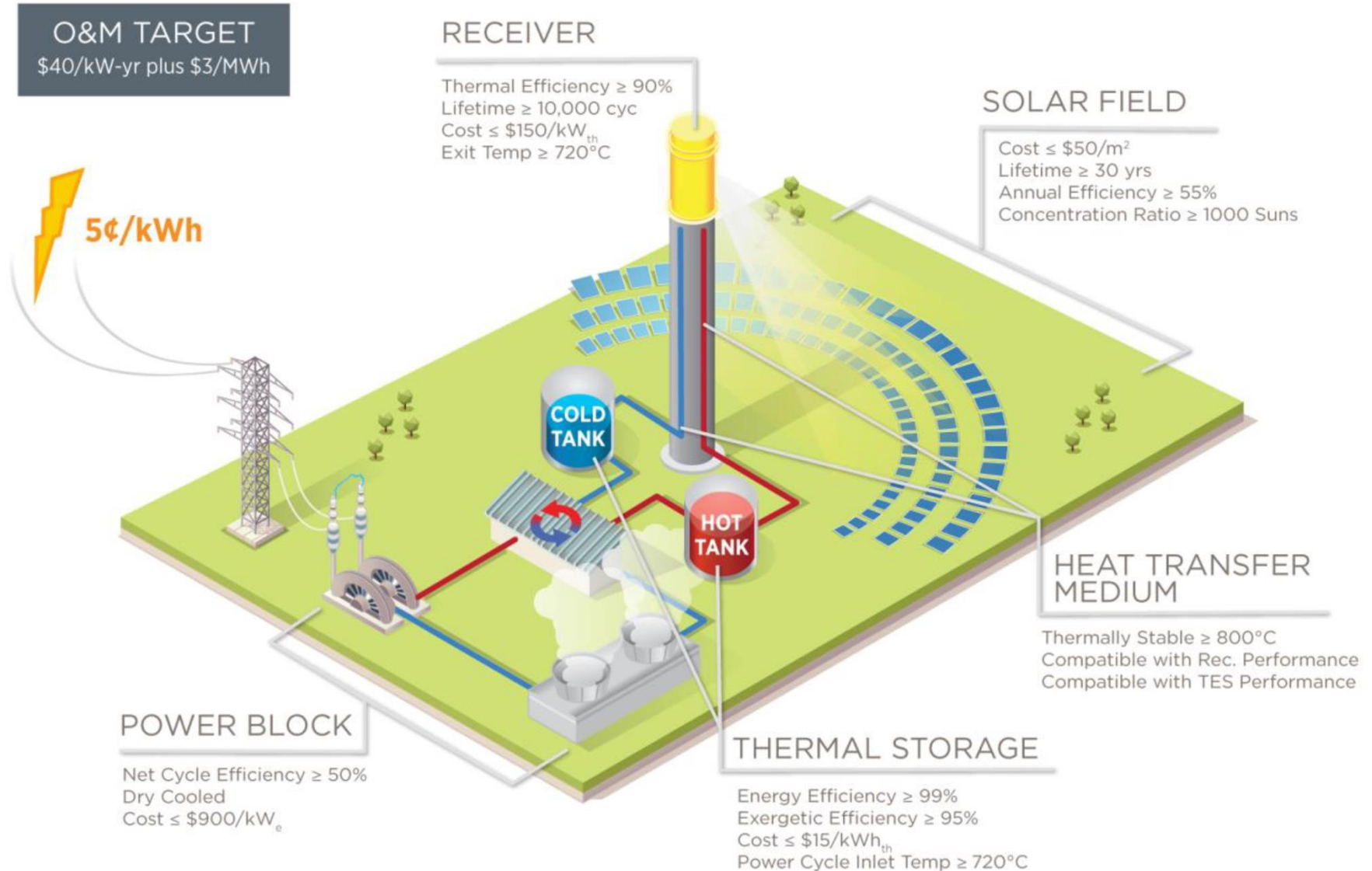
EE Interpretation of Each Stage

Stage	What Happens	Key EE Concept	Typical Quantities / Examples
Optics	Sunlight is collected and concentrated.	Energy concentration, geometry, radiative transfer, tracking	DNI (W/m ²), Concentration Ratio ($CR = \frac{A_{aperture}}{A_{receiver}}$)
Thermal	Concentrated light is absorbed and converted to heat.	Heat transfer (conduction, convection, radiation), thermodynamics	Heat transfer fluid (HTF) temperature, thermal power (MW _{th})
Turbine	Heat drives steam through a turbine to produce rotation.	Thermodynamic cycle (Rankine), entropy, enthalpy, efficiency	Steam pressure & temperature, turbine efficiency, mechanical power (MW)
Generator	Rotation is converted to electrical power.	Electromagnetic induction, AC generation, power electronics	Electrical power (MW _e), voltage, frequency (60 Hz), efficiency

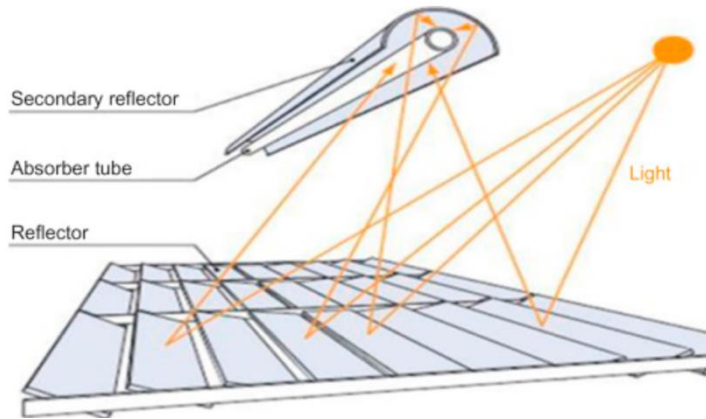


Overall System Efficiency = Product of Efficiencies at Each Stage – Losses
 Optical losses • Thermal losses • Turbine losses • Generator losses • Parasitic loads

CSP: Solar Tower



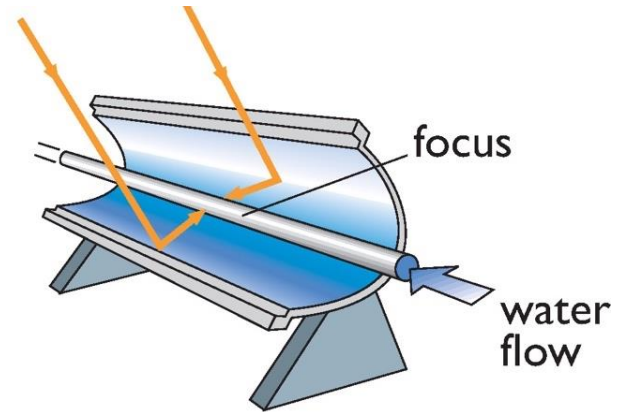
Solar Collectors



Linear Fresnel Reflector (LFR):

Uses rows of flat mirrors (Fresnel concept) arranged in straight lines to focus sunlight onto a linear receiver.

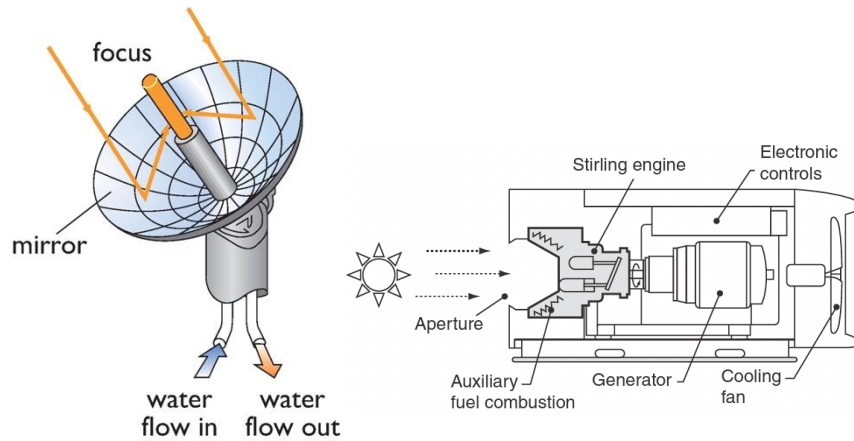
These mirrors are capable of concentrating the sun's energy to approximately 30 times its normal intensity.



Parabolic Trough Collector (PTC):

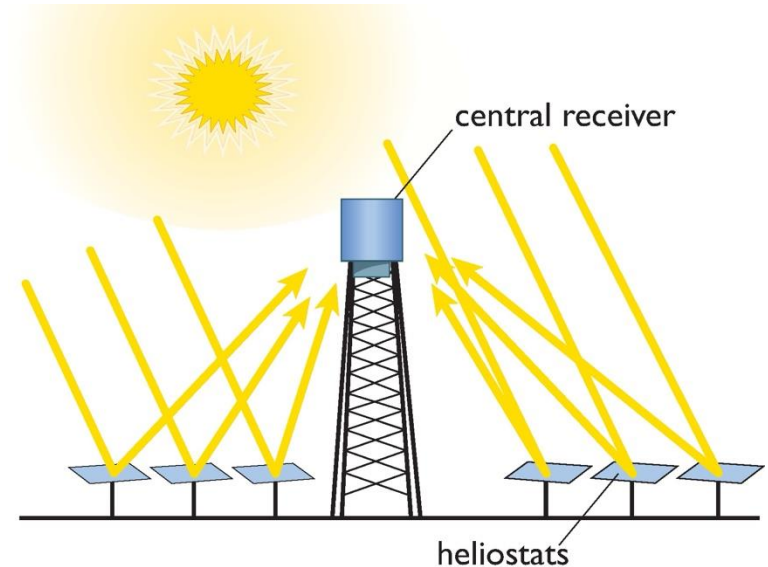
Rays are focused onto a pipe running down the center of a trough. The pipe carries a high temperature heat transfer fluid (e.g., mineral oil).

Solar Collectors (cont'd)



Parabolic Dish Collector (PDC):

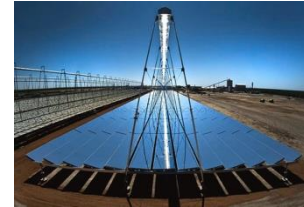
A large mirrored dish collects and concentrates the Sun's heat on to a receiver, which absorbs the heat and transfers it to fluid within a Stirling engine. The heat causes the fluid in the engine to expand against a piston or turbine.



Solar Power Tower (SPT):

The central receiver is a chamber where steam is produced directly used to drive a turbine. A heat transfer fluid is raised to a high temperature, to be pumped away to generate steam.

Technology Comparison



	Parabolic Trough	Solar Tower	Linear Fresnel	Dish-Stirling
Typical capacity (MW)	10-300	10-200	10-200	0.01-0.025
Maturity of technology	Commercially proven	Pilot commercial projects	Pilot projects	Demonstration projects
Key technology providers	Abengoa Solar, SolarMillennium, Sener Group, Acciona, Siemens, NextEra, ACS, SAMCA, etc.	Abengoa Solar, BrightSource, Energy, eSolar, SolarReserve, Torresol	Novatec Solar, Areva	
Technology development risk	Low	Medium	Medium	Medium
Operating temperature (°C)	350-550	250-565	390	550-750
Plant peak efficiency (%)	14-20	23-35*	18	30
Annual solar-to-electricity efficiency (net) (%)	11-16	7-20	13	12-25
Annual capacity factor (%)	25-28 (no TES) 29-43 (7h TES)	55 (10h TES)	22-24	25-28

* = upper limit is if the solar tower powers a combined cycle turbine.

Source: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2012/RE_Technologies_Cost_Analysis-CSP.pdf 32/46

Concentration Ratio (CR)

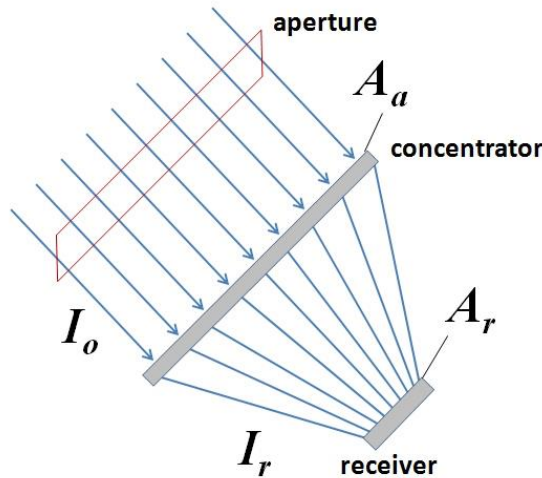


Fig: Schematic of solar light concentration

$$C_{geo} = \frac{\text{area of the aperture}}{\text{area of the receiver}} = \frac{A_a}{A_r}$$

- Purely based on areas
- Assumes perfect optics
- No losses
- Easy to compute

$$CR = \frac{\text{Power per unit area at focus}}{\text{Power per unit area at aperture}}$$

Based on energy, represents real performance. Includes:

- reflection losses
- tracking errors
- optical imperfections

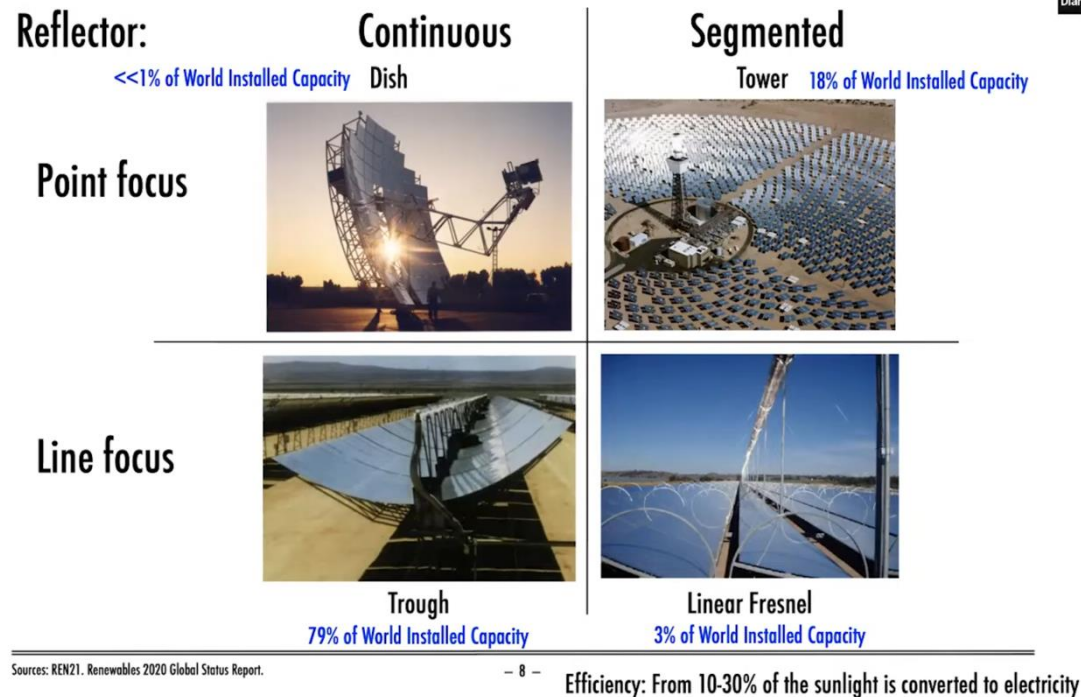
CR (cont'd)

$$CR \leq C_{geo}$$

C_{geo} is the ideal limit, while CR reflects the actual achieved concentration. They are typically of the same order of magnitude, but equality holds only under ideal conditions:

- perfect mirror
 - perfect alignment
 - no losses (never true in practice)
-
- CR typically ranges from 50–100, with some systems reaching ~1000.
 - Higher CR enables higher temperatures but requires more precise tracking and increases cost.

Reflector Classification



- **Line focus:** energy spread along a pipe (line)
- **Point focus:** energy concentrated onto a small spot
- **Segmented reflectors** approximate a shape using many pieces
- **Continuous reflectors** are optically equivalent to a smooth mirror (even if constructed from multiple pieces).

Continuous vs Segmented Reflectors

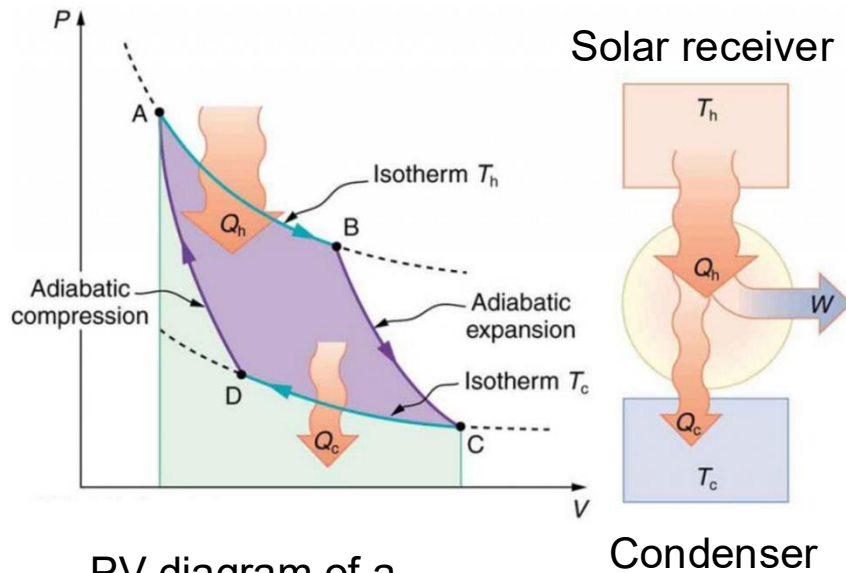
Feature	Continuous Reflector	Segment Reflector
Design	Single, unbroken reflective surface	Multiple flat or curved mirror segments
Optical Efficiency	Higher (due to smooth, uninterrupted surface)	Slightly lower (due to gaps and alignment tolerances)
Manufacturing	More complex and expensive	Easier and more cost-effective
Transportation	Difficult (large, rigid structure)	Easier (modular parts can be packed and shipped efficiently)
Installation	Requires precision, large handling equipment	Easier with smaller components
Maintenance	Harder—damage affects large sections	Easier—only affected segments need replacement

CSP Reflector Tradeoffs

Type	Temperature	CR	Tracking	Application
LFR	low	low	1-axis	cheap, low efficiency
PTC	medium	medium	1-axis	dominant tech
SPT	high	high	2-axis	utility-scale
PDC	very high	very high	2-axis	high efficiency, niche

Takeaway: Higher CR → higher temperature → higher efficiency
but higher cost and complexity

Thermal Efficiency = How Much Heat Becomes Work



PV diagram of a
Carnot cycle

- A heat engine is any system that converts heat $\rightarrow$ mechanical work
- Each point = a state of the system (pressure P , volume V)
- Moving along a curve = the gas is changing state (expanding or compressing)
- The purple area inside the loop = net work output W

$$\text{Thermal efficiency } \eta = \frac{\text{net work output}}{\text{total heat input}} = \frac{W}{Q_H} = \frac{Q_H - Q_C}{Q_H} = 1 - \frac{Q_C}{Q_H}$$

Carnot's Limit

Carnot's Theorem: $\frac{Q_C}{T_C} \geq \frac{Q_H}{T_H} \Rightarrow \eta_{\max} = 1 - \frac{T_C}{T_H}$

No real engine can exceed this limit. Inefficiency arises from:

- Carnot limit: efficiency is fundamentally bounded by T_H and T_C .
- Irreversibility: real processes (heat transfer, fluid flow) generate entropy, reducing useful work.
- Non-ideal losses: friction, leakage, and imperfect components.

Takeaway: All CSP design choices — mirrors, fluids, storage, cooling — are ultimately about managing losses relative to the Carnot limit.

- increasing T_H
- decreasing T_C

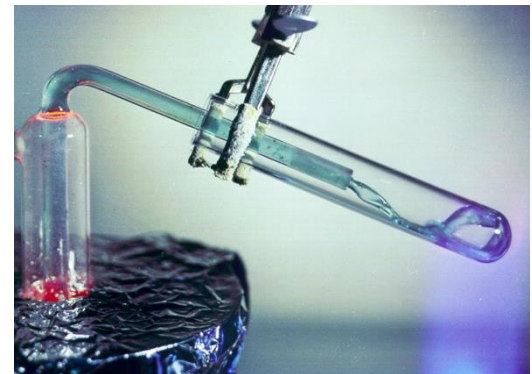
Heat Transfer Fluid (HTF)

- **HTF requirements:** high thermal stability, low viscosity, high boiling point, low freezing point, high heat capacity
- **Common HTFs:** synthetic oils, molten salts, water/steam
- **Oils vs salts:** oils for lower temperatures ($\leq 400^{\circ}\text{C}$); molten salts for higher temperatures ($\sim 600^{\circ}\text{C}$)
- **Water/steam:** high pressure at high T ; useful with direct steam cycles to avoid heat exchanger losses

Mineral oil: Any of various colorless, odorless, light mixtures of higher alkanes from a mineral source, particularly a distillate of petroleum



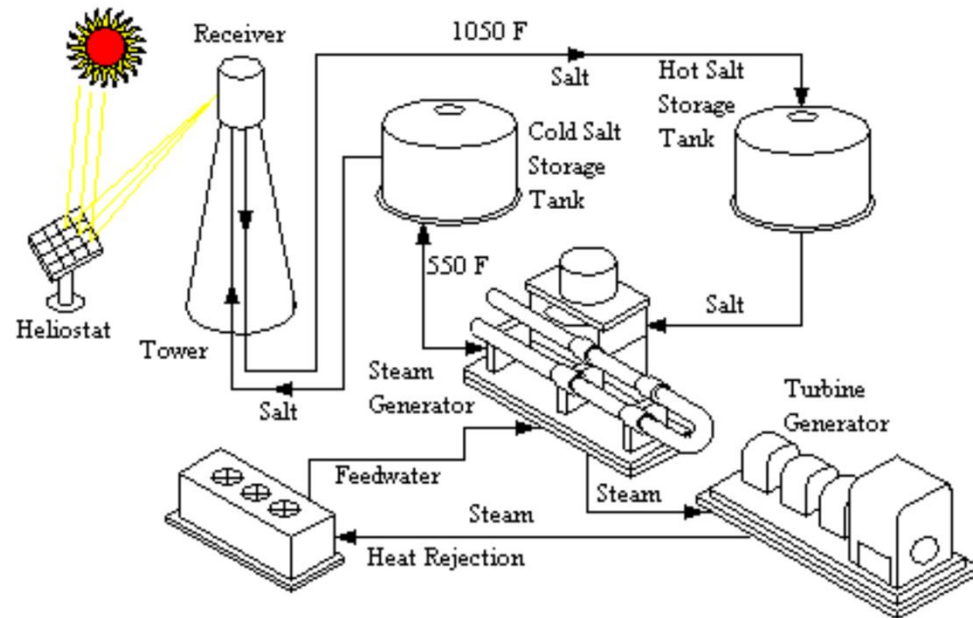
Molten salt: A mixture of nitrate salts. It is solid at standard temperature/pressure but enters the liquid phase due to elevated temperature



Thermal Energy Storage (TES)

Molten salts can be employed as a TES to retain thermal energy

- A commercially used technology to store the heat collected by CSP
- Non-solar apps: as a heat-transport fluid in chemical & metals industries



- 1) The salt melts at 268 °F & is kept liquid at 550 °F in an insulated cold storage tank
- 2) The liquid salt is pumped through panels in a solar collector where being heated to 1,050°F. Then sent to a hot storage tank
- 3) Pumped to a [steam generator](#) for driving a turbine/generator
- 4) A 100MW turbine needs a tank of about 30 ft tall and 79 ft in diameter to drive it for 4 hours

Advantages of Molten Salts

Molten salts offer several advantages over high temperature oils:

- Allow for higher temperatures that reduces the volume of storage.
- More environmentally friendly: cheaper, nonflammable, nontoxic.
- Highly efficient: potential round-trip charging/discharging cycle efficiencies over 98% and standby thermal losses of about 0.03%/h.

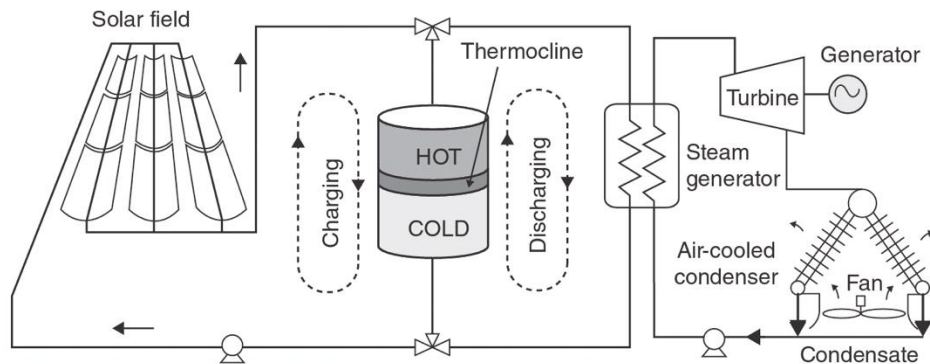
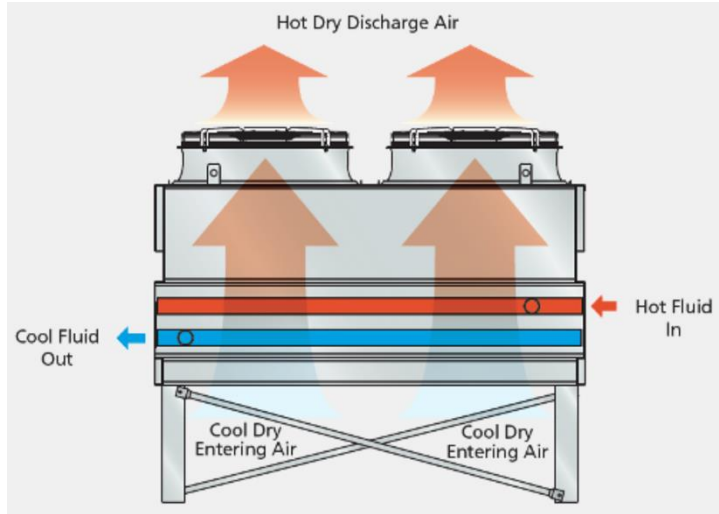


FIGURE 8.8 A direct, single-tank, thermocline, molten salt storage system.

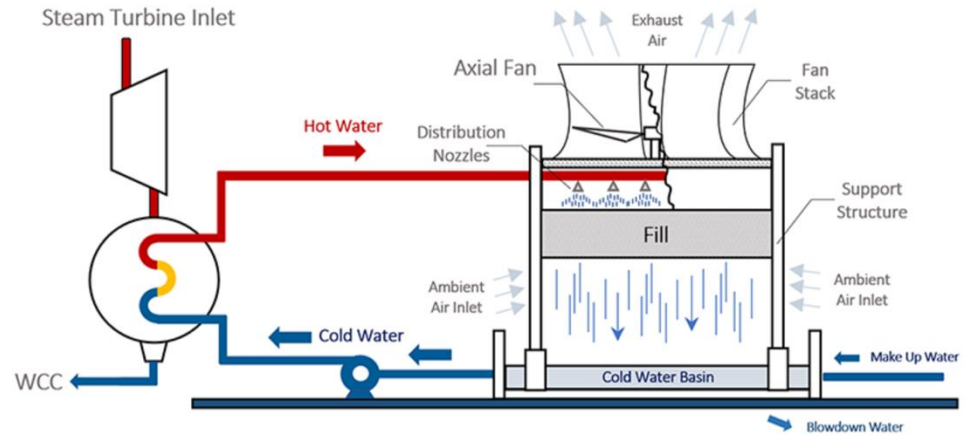
Fig: Single-tank thermocline storage. Besides molten salt, a low-cost filler material such as quartzite rock in combination with silica sand would occupy much of the volume (to inhibit vertical mixing of the layers while reduce the fluid volume required).

Types of Cooling Systems

Dry cooling



Wet cooling



The main goal of cooling towers: Remove the excess heat and reject it to the outside environment.

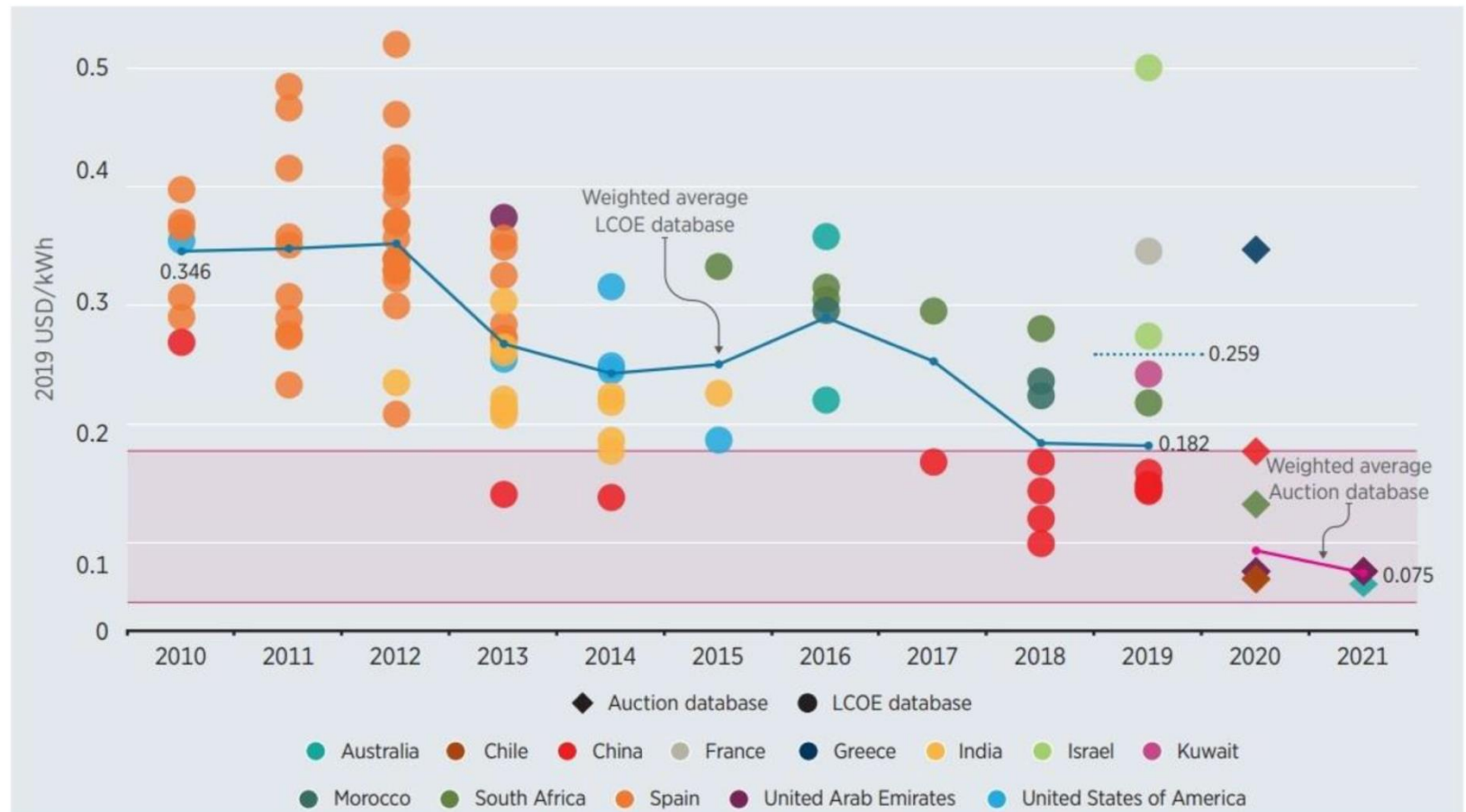
Two main cooling towers:

- Closed-loop (dry): fluid stays in pipes, no direct contact with air
- Open-loop (wet): fluid contacts air, some water evaporates

Tradeoff Between the Two-type Cooling Systems

Feature / Aspect	Dry Cooling in CSP	Wet Cooling in CSP
Cooling Medium	Air (ambient)	Water (evaporation-based)
Water Consumption	Minimal (used only for cleaning, not cooling)	High (due to evaporation losses)
Cooling Efficiency	Lower, especially during hot days	Higher, effective even in high ambient temperatures
Thermal Cycle Efficiency	Reduced due to higher condenser temperature	Higher due to lower condenser backpressure
Capital Cost	Higher (due to large air-cooled condenser units)	Lower (simpler cooling tower design)
Operating Cost	Lower (minimal water treatment & procurement)	Higher (cost of water procurement and treatment)

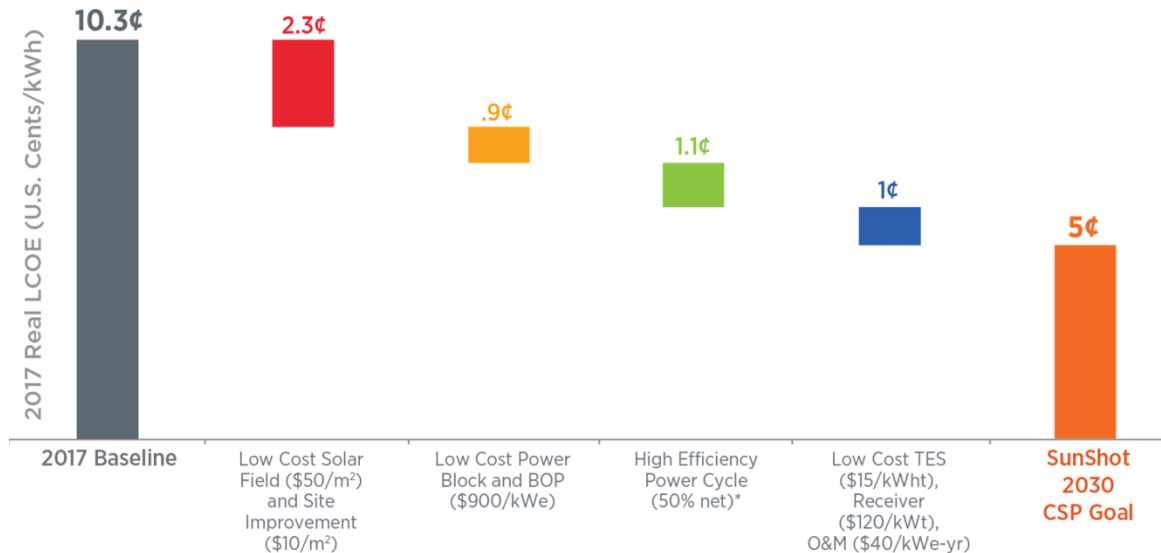
CSP LCOE



- **LCOE database:** Provide cost-related data to help investors and developers make informed decisions.
- **Auction database:** Provide information about renewable energy projects that have been awarded contracts through auctions.

CSP Requires Interdisciplinary Solutions

A Pathway to \$0.05 per kWh for Baseload CSP



*Assumes a gross to net conversion factor of 0.9



Fig: Two Power Tower plants in southern Spain (2009)

There is **no single CSP challenge** but rather **a series of challenges** of:

- heat transfer, fluid mechanics, thermodynamics,
- optical physics, materials science, extreme automation in the solar field, corrosion mitigation, advanced manufacturing, thermo-mechanical engineering design, low-cost sensors & control, and
- predictive operations and maintenance, among others.